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Collisions on Feistel-MiMC and univariate GMiMC

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Abstract. MiMC and GMiMC are families of MPC-friendly block ciphers and hash functions. In this note, we show that the block ciphers MiMC- $2n/n$ (or Feistel-MiMC) and univariate GMiMC are vulnerable to an attack which allows a key recovery in $2^{n/2}$ operations. This attack, which is reminiscent of a slide attack, only relies on their weak key schedules, and is independent of the round function (x^3 here) and the number of rounds.

Keywords: MiMC, MPC, symmetric cryptanalysis

1 Description of the ciphers

1.1 MiMC- $2n/n$

MiMC- $2n/n$ [Alb+16] is a $2n$ -bit block size, n -bit key block cipher. It claimed n bits of security. Its round function is described in Figure 1, and can be written as

$$R_k^i(x_L, x_R) = x_R \oplus (x_L \oplus k \oplus c_i)^3, x_L \quad .$$

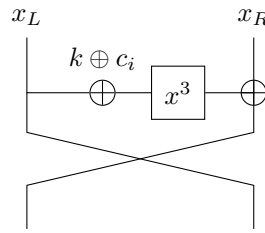


Figure 1: MiMC- $2n/n$ round function

1.2 GMiMC

GMiMC [Alb+19] generalizes the MiMC- $2n/n$ construction to generalized Feistels. Two key schedules are proposed. The univariate key schedule uses a fixed key for each round, while the multivariate key schedule uses t initial keys and updates the round keys. Their claimed security corresponds to the number of bits of the key. Four generalized feistel constructions are proposed:

GMiMC-crf. GMiMC-crf has t branches and adds a function of $t - 1$ branches on one branch. The round function is

$$R_k^i(x_1, \dots, x_t) = x_2, \dots, x_t, x_1 \oplus \left(\bigoplus_{j=2}^t x_j \oplus k \oplus c_i \right)^3.$$

GMiMC-erf. GMiMC-erf has t branches, and adds a function of one branch on all the other. The round function is

$$R_k^i(x_1, \dots, x_t) = x_2 \oplus (x_1 \oplus k \oplus c_i)^3, \dots, x_t \oplus (x_1 \oplus k \oplus c_i)^3, x_1.$$

GMiMC-Nyb. GMiMC-Nyb has $2t$ branches, and adds a function of each odd branch to the next branch. The round function is

$$R_k^i(x_1, \dots, x_t) = x_2 \oplus (x_1 \oplus k \oplus c_{ti})^3, x_3, x_4 \oplus (x_3 \oplus k \oplus c_{ti+1})^3, \dots, x_{2t} \oplus (x_{2t-1} \oplus k \oplus c_{ti+t-1})^3, x_1.$$

GMiMC-mrf. GMiMC-mrf is a generalization of the previous construction with a permutation of the branches that change for each round.

2 Attacks

2.1 Attack on MiMC-2n/n

The attack relies on an invariant property of the round function, and can be seen as a slight generalization of a slide attack presented in [BNPS19]. The invariant property is described in [Figure 2](#).

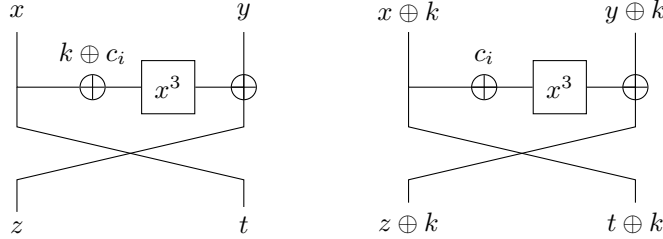


Figure 2: Illustration of [Lemma 1](#)

Lemma 1. Let R_k^i be the round function of MiMC-2n/n with the key k for round i . Then for all x, y, k, i , $R_k^i(x, y) \oplus (k, k) = R_0^i(x \oplus k, y \oplus k)$

Proof. $R_0^i(x \oplus k, y \oplus k) = (y \oplus k \oplus (x \oplus k \oplus c_i)^3, x \oplus k) = (y \oplus (x \oplus k \oplus c_i)^3, x) \oplus (k, k) = R_k^i(x, y) \oplus (k, k)$ $\square$

Theorem 1. Let E_k be MiMC-2n/n with the key k . Then, for all x, y, k , $E_k(x, y) \oplus (k, k) = E_0(x \oplus k, y \oplus k)$.

Proof. By induction over the number of rounds. The base case is [Lemma 1](#). If the property holds after $i - 1$ rounds, then

$$(R_k^{i-1} \circ R_k^{i-2} \dots \circ R_k^1)(x, y) \oplus (k, k) = (R_0^{i-1} \circ R_0^{i-2} \dots \circ R_0^1)(x \oplus k, y \oplus k).$$

By Lemma 1,

$$\begin{aligned} (R_0^i \circ R_0^{i-1} \cdots \circ R_0^1)(x \oplus k, y \oplus k) &= R_0^i((R_0^{i-1} \circ R_0^{i-2} \cdots \circ R_0^1)(x \oplus k, y \oplus k)) \\ &= R_0^i((R_k^{i-1} \circ R_k^{i-2} \cdots \circ R_k^1)(x, y) \oplus (k, k)) = (R_k^i \circ R_k^{i-1} \cdots \circ R_k^1)(x, y) \oplus (k, k) \quad \square \end{aligned}$$

Corollary 1. *Let E_k be MiMC- $2n/n$ with the key k . Let $f(x) = E_k(x, x) \oplus (x, x)$ and $g(x) = E_0(x, x) \oplus (x, x)$. Then $f(x) = g(x \oplus k)$.*

Proof.

$$\begin{aligned} g(x \oplus k) &= E_0(x \oplus k, x \oplus k) \oplus (x \oplus k, x \oplus k) = E_k(x, x) \oplus (k, k) \oplus (x \oplus k, x \oplus k) \\ &= E_k(x, x) \oplus (x, x) = f(x) \quad \square \end{aligned}$$

Key recovery. The key recovery simply consists in looking for a collision between f and g from Corollary 1, which can be done in time $2^{n/2}$ as the two functions have an n -bit input. This contradicts the claim of n bits of security of MiMC- $2n/n$.

Hash function. MiMC can be used keyless as a permutation for a sponge-based hash function. As there is no key in this construction, it is unclear how Theorem 1 could be used to attack the hash function.

2.2 Attacks on GMiMC

In most cases, the same property can be found in univariate GMiMC, that is, $E_k(x_1, \dots, x_t) \oplus (k, \dots, k) = E_0(x_1 \oplus k, \dots, x_t \oplus k)$, which allows to apply the same attack as in the MiMC- $2n/n$ case.

GMiMC-Nyb and GMiMC-mrf. One round of GMiMC-Nyb and GMiMC-mrf can be seen, up to a permutation of the branches, as t Feistel in parallel. Hence, the property holds.

GMiMC-erf. The added function only depends on one input branch, hence the property also holds.

GMiMC-crf. The function is slightly different in that case, as it depends on more than one branch. For the property to hold, we must have that

$$((\oplus_{j=2}^t x_j) \oplus k \oplus c_i)^3 = (\oplus_{j=2}^t (x_j \oplus k) \oplus c_i)^3.$$

Hence, the property holds only if t is even.

2.3 Variants in large characteristics

MiMC and GMiMC can also be defined over a finite field of large characteristic. In that case, the property we have is $E_k(x_1, \dots, x_t) + (k, \dots, k) = E_0(x_1 + k, \dots, x_t + k)$, and the same attack can be applied. The only exception is GMiMC-crf, where we need to have $k + k = 0$ for the property to hold.

2.4 Quantum attacks

The collision property corresponds to a hidden period, and as such, permits a key recovery in $\mathcal{O}(n)$ quantum queries. With a restriction to classical queries, these attacks happens to be in a form suitable for the offline Simon's algorithm [Bon+19], which allows to make a key recovery in $\mathcal{O}(2^{n/3})$ classical queries and quantum time.

3 Conclusion

We have shown that MiMC- $2n/n$ and all the versions of univariate GMiMC except some instances of GMiMC-crf are vulnerable to a collision attack. More generally, this demonstrates that using round constants is not enough for a key schedule to secure a Feistel or generalized Feistel construction.

This attack does not appear to be applicable to the other MiMC construction, MiMC- n/n , nor to the hash functions based on any version of MiMC or GMiMC.

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